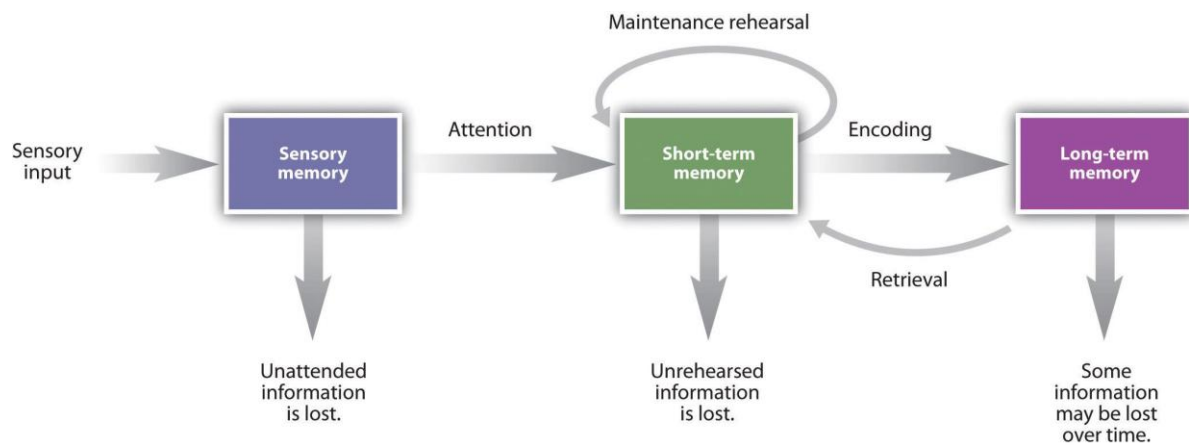


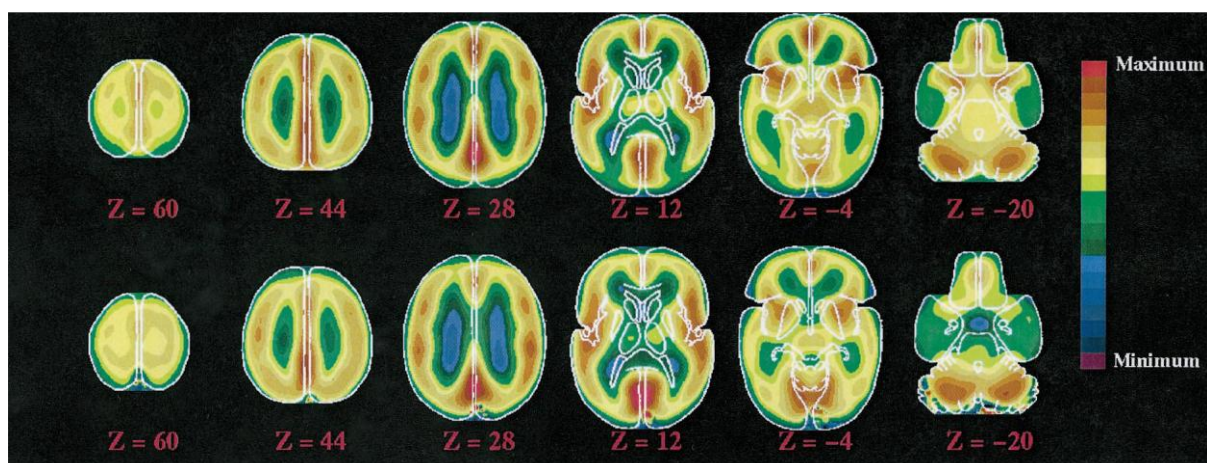


The brain at rest

We talk about how the brain works at rest because it has several implications for the study of the brain. Typically, we have a contrast in which we have a period of rest and a period in which we do a specific task, so we expect the regions that react to the specific stimulation to show a change in activity and then we compare the activity that we observe during the task to the images of brain at rest. So, the regions that show a response during stimulation are activated during it, involved in the task, and what we consider are the temporal and spatial correlates of the activation. What we were able to see is that while doing a task there are also regions that have a negative response, some parts get deactivated, and it may mean that during rest there are parts of the brain strongly activated. This led to the creation of one of the most widely used file dime in research with f-MRI that is called resting state, the most common type of data (simple way to detect it).

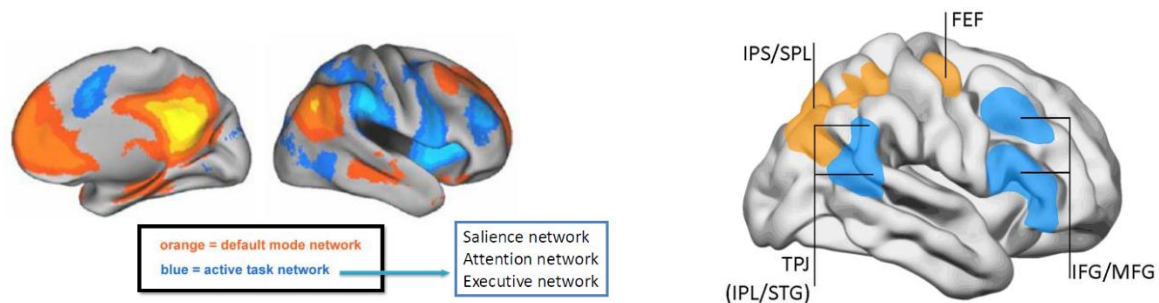
The idea at the very beginning was to collect a lot of data and see what would change in the brain for a pathological condition, and maybe even learn how to classify brains. It didn't really work out, because alterations in the brain can be found but are never specific in practice, we can have the same alterations in many pathological conditions.

There are two types of resting state, eyes open or eyes closed. The first one is typically preferred because you may fall asleep with eyes closed. The common view was that the resting state was a condition with minimal brain activity, but now the view is changed, and has been found that the increment in brain activity because of a task, it's very small (5% change in metabolic request), so even at rest the brain has a large energy consumption.



Quantitative maps of blood flow (Upper) and oxygen consumption (Lower) during quiet rest with eyes closed. Note the large variation in blood flow and oxygen consumption across regions of the brain. These vary most widely between grey and white matter. Despite this variation, blood flow and oxygen consumption are closely matched. There is a network of areas that are involved, and it is called “default mode network” because it works when you are at rest. These network includes the medial pre-frontal cortex, the medial parietal cortex, some parts on the sides, the superior frontal cortex, the lateral parietal cortex and also the lateral temporal cortex.

People started to do a lot of experiments to focus on what changed, also in negative terms, and they created some types of maps, representing the contrast between the task and the rest condition.

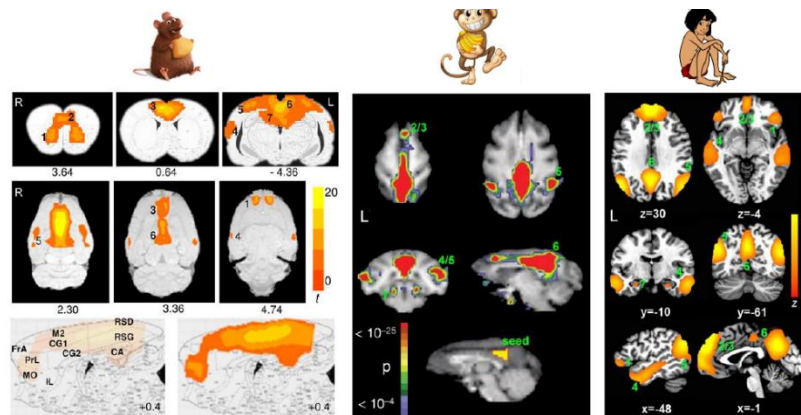


These types of images led to a view in which there are two main systems in the brain, one is the default mode network, and the other is the attention network (task positive network). In different tasks the patient will have activated some regions that are important for attention (right picture) toward the external world, top-down and bottom-up attention. In orange we have the dorsal system of the attention network, which includes the intraparietal sulcus (IPS) and the frontal eye field (FEF), that controls the movement of the eyes and so the direction of the attention, this is the voluntary top-down network for attention. The ventral system (in blue) includes indeed the right temporal parietal junction and the right ventral frontal cortex, the inferior frontal gyrus (IGF) and the medial frontal gyrus (MFG), and this is a system that activates when the attention is recalled spontaneously towards something. The moment you make your patient pay attention to something; the default mode network will decrease its activity.

The DMN oscillates spontaneously in activity every about ten-twenty seconds (increases and decreases, never 0). During the resting state the patient starts thinking about a lot of things, just like daydreaming, envisioning the future, retrieving memories, and gauging others' perspectives, and it appeared that DMN may be related to this process of thinking. Then they started to experiment what is called experience sampling (you keep the patient in the f-MRI and at a random moment you send an alarm and ask the patient what they felt, while scanning), so that you can have a report of what the patient was thinking about. It seems that this set of regions is related to something that is directed internally, a processing of internal experiences, everything that is unrelated to what happen around us. So, the moment people stop reasoning internally and pay attention to the external world the DMN is deactivated.

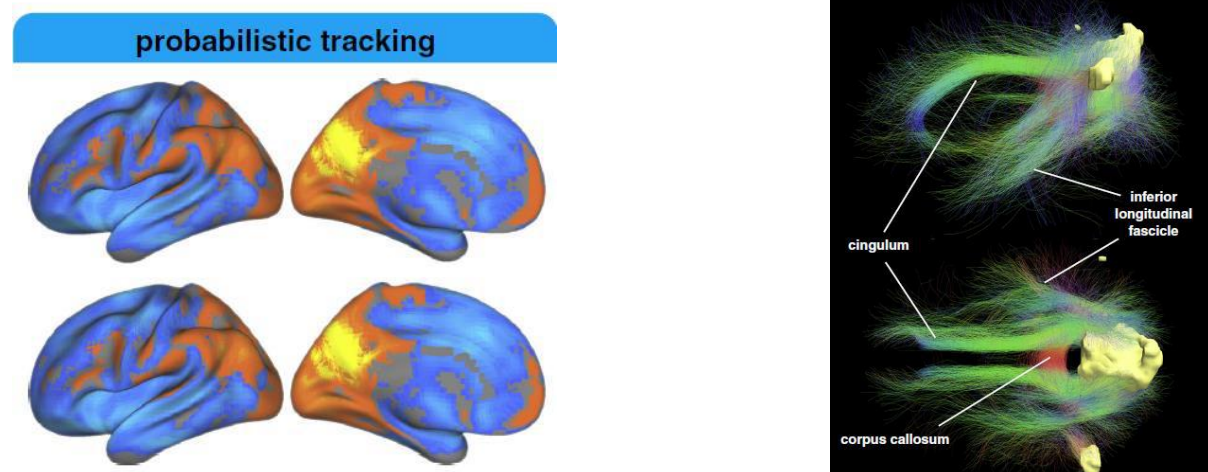
The function of the DMN is still unclear, and its role is still debated. The main school of thinking says that it is related to introspection and self-referential thoughts, when the direction is directed internally, spontaneous oscillation and continuous conflict between internal and external directing of thoughts. Some studies suggested indeed that the oscillation of DMN activity can influence our ability to pay attention to external elements, if I present you a stimulus in a moment in which the DMN is strongly activated it's very likely that you will miss it.

It's normal for the brain to shift from a task to another, but multitasking in the sense that we think about it like something in which our consciousness is divided across multiple things does not exist, we can only focus on one thing at a time. And here is the same thing, we can only focus internally or externally, not both at the same time.



Rats and Monkeys possess a DMN that is broadly similar to the DMNs of humans. A well-organized, intrinsically coherent DMN appears to be a fundamental feature in the mammalian brain.

What is interesting about DMN is also the fact that it seems to be a sort of a backbone for the brain, meaning that if you look to the structural connectivity, you see that they the strongest connected regions in the brain. Tractography is a technique (particular application of f-MRI) in which you use a signal that is sensitive to water movement in the brain, so water can move inside the cells only along the axons and if you look at the direction of the dominant movement in a specific voxel in the brain, you can track fibres of axons of the neurons.



The green things are all connections, and the region in yellow is the posterior part of the default mode network (praeuneus). As we can see there are strong connections between this area and the frontal region, the anterior part of DMN. These are the strongest connections in our brain. In children apparently is not exactly the same, the activity is not in the same regions than in adults, the visual cortex is strongly activated, and also in the motor cortex but not in the other regions, contrary to adults. In children most of the connections are local, in adults there are many long-range connections, and then during the growth process there is a reorganization of the brain.

The DMN is mainly studied with f-MRI.

One type of study is based on the view of DMN function depending on the condition of the patient or whether the DMN changes depending on the state of the patient. Even if you ask to meditate there is a change in DMN activity. Drugs and sleep also affect the DMN, during sleep particularly it gets stronger. Many different activities can have effects on the connectivity of DMN.

- **Attention to external stimuli:** A shift of attention toward specific stimuli or tasks is associated with a reduced connectivity within the DMN.
- **Meditation:** Long-term meditation practitioners may show reduced activation and reduced functional connectivity in the DMN.
- **Psychedelic drugs:** Psilocybin has been shown to be associated with a reduction of the blood flow to areas of the DMN (posterior cingulate cortex and medial prefrontal cortex).
- **Sleep:** Sleep deepening is associated with a decrease in connectivity between the posterior cingulate cortex and medial prefrontal cortex.
- **Sleep deprivation:** Sleep deprivation results in a decrease in connectivity within the DMN. A decreased anti-correlation between the DMN and the TPN can be also observed.

It's impossible that one f-MRI scan can determine if I have a pathological condition, it's not specific. The DMN overlaps areas involved in major brain disorders. In fact, alterations of the DMN have been observed in several pathological conditions. Discerning precisely which aspects of the network are affected by different disorders may provide new diagnostic tools. For example:

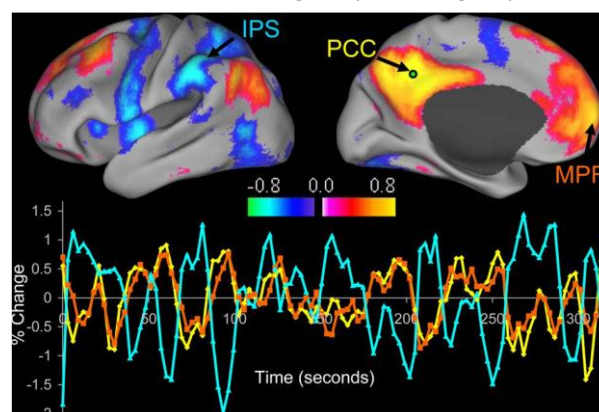
- **Alzheimer's Disease:** Brain areas that show marked atrophy in Alzheimer's overlap very closely with major centres of the DMN.
- **Depression:** Hyperconnectivity in the DMN seems associated with rumination (always thinking about bad things) in Depression.
- **Autism:** Patients may fail to reduce DMN activity during cognitively demanding tasks.

The methods used to study resting state functional MRI data, are mainly two (map-based methods):

- Seed-based functional connectivity (FC) analysis, hypothesis driven method (bottom-up).
- Independent component analysis (ICA), data driven method (top-down).

But also, a graph-based method, considering connectivity matrices and graph analysis.

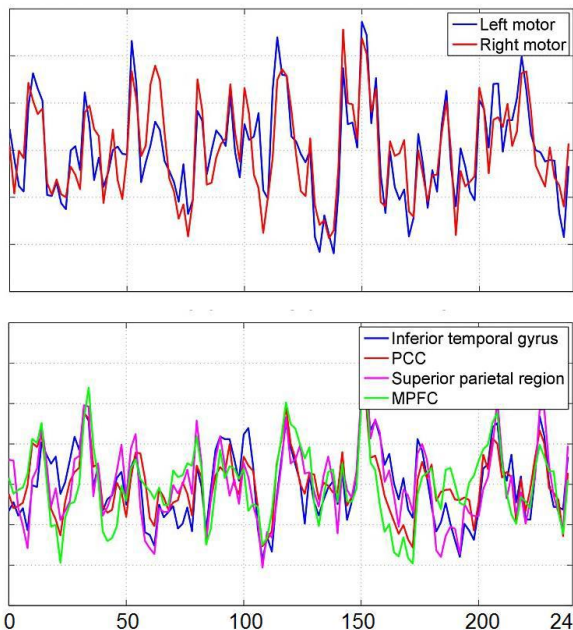
In seed-based correlation mapping you just need to decide a seed region, a target region that you use to look in the brain for other regions that correlate with the seed. I take a group of voxels in the brain, and I extract the time-series of the signal (f-MRI signal).



Example.

In this example we can see the region in yellow (PCC), which has a signal in yellow (BOLD signal) that records the hemodynamic changes over-time, through f-MRI. Then I take this time-series, and I calculate an instantaneous correlation (Pearson correlation coefficients) between this region and all the other processes in the brain. The correlation coefficient is close to 0 if there is no similarity at all, 1 if the time-series are identical, -1 if they go in opposite directions. Finally, you can apply some statistical thresholding and what you get is a map in which you can see in yellow or orange the regions correlated to your seed and in blue the regions that show an anti-correlation. Looking at this is clear that you have in orange the default mode network and in blue the regions that correspond to the attention network.

This kind of approach is called functional connectivity, that historically has been used to distinguish between the instantaneous correlation and lagged correlation. Basically, is used in contrast to the other one that is effective connectivity, that was originally (before 2015) used to indicate causal correlation. The idea was that seeing a change in activity in a region and then a corresponding change in activity in another region but after some time that would mean that the first region was causing the change to the secondary. But we cannot say this, in fact now effective connectivity is just used to indicate lagged connectivity.



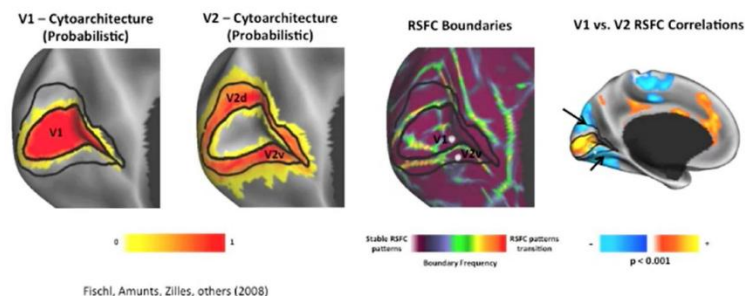
Then researchers realized something. If you place the seed in another part of the brain that is not part of DMN, you see that other networks pop-up. For example, if you place the seed (red one) on the right motor cortex and you explore the correlation between this time series and all other voxels in the brain, you see that at some point you will find regions that show similarity in activity. In the case at the left, for example the left motor cortex. They communicate even if the subject is not doing anything, only because they are strongly connected.

Hence, using this type of approach you can explore multiple networks in the brain, just moving the seed around the brain.

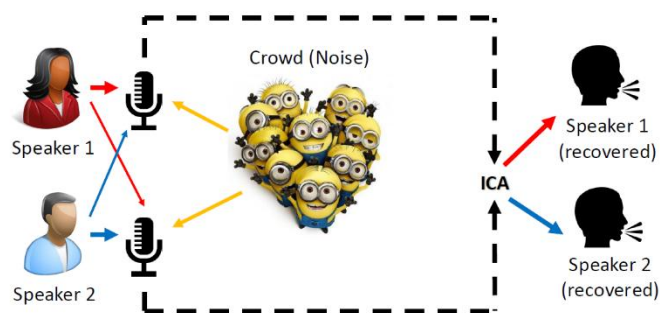
If you start to move around you can identify multiple networks and points for which if you move outside the area you start to see that the network changes, in this way you can map which part of the brain belong to which network and create functional atlases of the brain. These atlases are quietly available now to download and there are atlases that include 1000 regions, segmentation of very small regions, or 7 big areas, identification of larger networks.

The regions that you see at rest are exactly showing the same type of interaction that they show when they actually interact actively, so when you do this kind of approach in the visual cortex, and you start to move around within the visual area V1, when you move outside the border of this region you start to see a change in connectivity and you can move around and see that it shows a pattern of connectivity that reflects the secondary visual cortex (V2d, V2v). So basically these are

regions that have a specific functional meaning in terms of brain function, the same has been seen for the motor cortex or even in people with the eyes closed and blind people, especially for this last case you can see moving around in the visual cortex that the organization of it is still partially preserved.



The independent component analysis is a data driven approach, a method for separating multivariate signals into subcomponents.

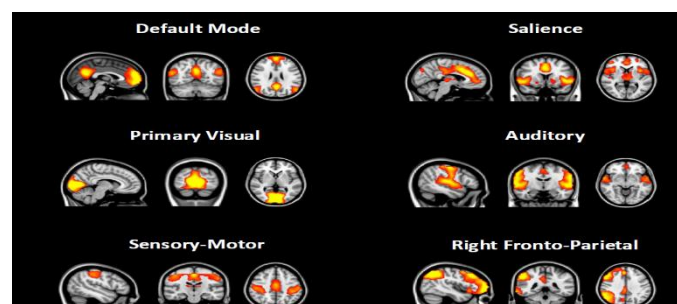


In this example you have the two speakers and some noise. What you recover in the end is that the signal sums up in the sensors (microphone), and with this method you can separate the sources assuming that they are independent and that each source has a non-gaussian distribution (blind-source separation).

This can be done also with f-MRI, but is not exactly optimal for this method, except for certain adaptations of the algorithm. The reason is that f-MRI has very low temporal resolution, so even if you acquire 10 minutes of data you get few 100s time points, considering that on the spatial site you have several thousands voxels, this disparity is not optimal.

ICA separates a signal into non-overlapping spatial and time components and for this reason can be used in the detection of resting state networks. It is highly data-driven and allows for better removal of noisy components of the signal (motion, scanner drift, etc). It also has been shown to reliably extract default mode network as well as many other networks with very high consistency.

Through this approach you are able to detect noise, just like head motion, heart pulsation, respiration, but they also found several networks. The most dominate one is obviously the default mode network, but you can separate for example a network that corresponds to the visual cortex (you have bilaterally the two visual regions), the sensory motor network (with the two motor cortices), the auditory network (involvement of the temporal cortex).



These networks can be studied in terms of not only functioning but also anatomical connections, because the functioning reflects the anatomy of the brain. If you do tractography (able to track the direction of axons), you realize that these functional networks actually correspond very well to anatomical connections in the brain. Sometimes you can find regions that show strong functional connectivity even though they are not anatomical connected.

By using this approach several networks have been identified, and they are independent from each other.



Resting-state networks consist of anatomically separated but functionally connected regions displaying a high level of correlated BOLD signal activity. These networks are found to be quite consistent across studies, despite differences in the data acquisition and analysis techniques. Most of these resting-state components represent functional networks of regions that are known to share and support specific cognitive functions. This organization can be affected in different ways by neurological disorders and brain lesions; the grade of damage can be seen by comparing the signal in the lesioned person to an atlas using mathematical transformations that can modify the shape of the brain that may be changed by the disorder.

One approach used for processing data is taking the anatomical and functional images and putting data in the space of the atlas (standard space, average brain anatomy), and then you apply brain parcellation to the data, then you extract the average time series for each of the regions of interest that you want to analyse so that finally you can use this data for example to calculate how much the activity of the DMN is similar or different to the activity of another network. You can also calculate matrices of correlation, calculating for each region the correlation with all the other regions of the brain, in which the diagonal corresponds to the identity (connectivity between the region and itself), and in each row you can see the connectivity between the region and all the other regions in the brain. Once you have this kind of matrix you can perform different types of analysis, for example you can build some graphs that represent connectivity within the brain.



In this case each dot represents a region of the brain, and the lines represent the strength of the connectivity (correlation between them). Once you have a graph like this you can extract many parameters.

- **Global efficiency:** Is a measure of how efficiently a network exchanges information. It is the average inverse shortest path length in the network and is inversely related to the characteristic path length.
- **Clustering coefficient:** Is a measure of the degree to which nodes in a graph tend to cluster together. It is the fraction of triangles around a node and is equivalent to the fraction of node's neighbours that are neighbours of each other.
- **Characteristic path length:** the average shortest path length in the network (lengths of shortest paths between all pairs of nodes).